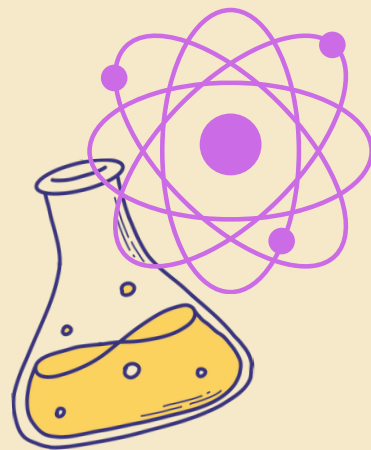


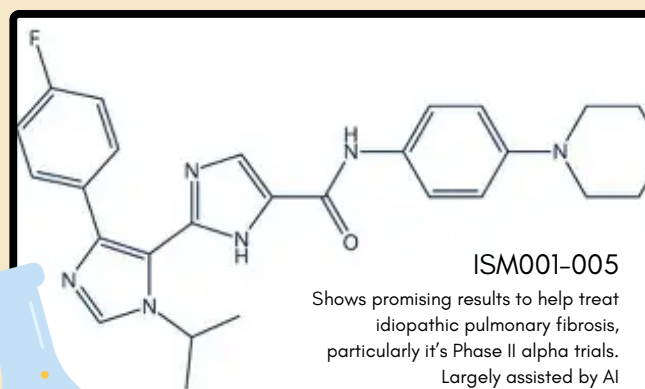
Artificial Intelligence: Organic Synthesis

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Background info

Organic synthesis of very large molecules like drugs is very tedious due to many considerations. AI can help to make organic synthesis more economical, efficient and in a manner that mitigates many of these existing concerns.



Strategy of AI

Molecule Screening

AI is able to map molecules by tagging them based on its carbon skeleton and functional groups which aids in discovering compounds that effectively target the desired protein

Retrosynthesis

AI can **work backwards** from the product molecule to break it down into starting materials.

Rule based: uses classic rules of reactivity
VS

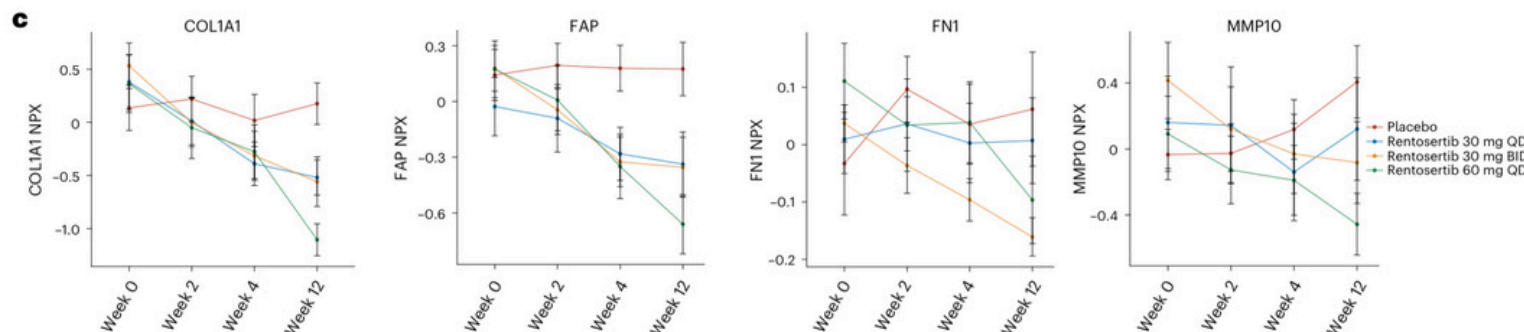
Rule free: treating organic chemistry as a language

Testing Reactivity

AI harnesses the power of techniques like density field theory, using **quantum physical theory** to predict **reactivity** and **selectivity**

Drug Design

AI simulates chemical conditions **in-vivo** and **screens** proteins to target



ISM001-005 has been shown to effectively downregulate inflammatory markers.

Conclusion

AI is starting to **revolutionise** the synthesis of organic molecules, giving more **hope** of treating diseases currently considered incurable.

